Lecture 13 - April 24 - 2026

11.9 Representation of Functions as Power Series.

General idea:

$\sum_{n=0}^{\infty} c_n (x-a)^n$ (power series.) $\longrightarrow$ $f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$ we can define Dom f.

If $f(x)$ is a function, we know Dom f. $\longrightarrow$ what is its power series?

this is important b/c, $f(x) = \frac{\sin x}{x}$; $\text{Dom } f = \mathbb{R} - \{0\}$

- $f'(x) = \frac{d}{dx}(\sin x \cdot x^{-1}) = \frac{\cos x}{x} - \frac{\sin x}{x^2} = \frac{x\cos x - \sin x}{x^2}$
- $\int \frac{\sin x}{x} dx = DNE$ i.e., the there is not a function $F(x)$ s.t.

$$F'(x) = \frac{\sin x}{x}.$$

The only way is try to approx $\frac{\sin x}{x}$ by a power series.

$$\frac{\sin x}{x} = \sum_{n=0}^{\infty} c_n x^n$$

$$\int_1^2 \frac{\sin x}{x} dx \approx \int_1^2 \left(\sum_{n=0}^{\infty} c_n x^n\right) dx$$

$$= \sum_{n=0}^{\infty} \int_1^2 c_n x^n dx$$

$$= \sum_{n=0}^{\infty} c_n \int_1^2 x^n dx$$

$$= \sum_{n=0}^{\infty} c_n \left.\frac{x^{n+1}}{n+1}\right|_1^2$$

$$= \sum_{n=0}^{\infty} c_n \left(\frac{2^{n+1}}{n+1} - \frac{1}{n+1}\right).$$

Representation of Functions using Geometric Series.

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots = \sum_{n=0}^{\infty} x^n \quad ; \quad |x| < 1$$

Example: Express $1/(1+x^2)$ as a power series and find the interval of convergence.

Sln.

$$\frac{1}{1-y} = \sum_{n=0}^{\infty} y^n \quad ; \quad |y| < 1 \qquad \text{let } y = -x^2$$

$$\frac{1}{1-(-x^2)} = \frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n}$$

let $a_n = (-1)^n x^{2n}$

$$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \sqrt[n]{|x|^{2n}} = |x|^2 = x^2 < 1 \qquad \text{So, } x \in (-1, 1).$$

if $x = -1$, then $\sum_{n=0}^{\infty} (-1)^n (1)^{2n}$ which is divergent.

if $x = 1$ then $\sum_{n=0}^{\infty} (-1)^n (1)^{2n}$ which is divergent.

The easy way: We have $|y| < 1$ then $y = -x^2$ $\quad |-x^2| < 1 \iff x^2 < 1$.

So, $x \in (-1, 1)$.

Example: Find a power series representation for $1/(x+2)$.

Sln. Using:

$$\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n \quad ; \quad |x| < 1$$

$$\frac{1}{2+x} = \frac{1}{2\left(1+\frac{x}{2}\right)} = \frac{1}{2\left(1-\left(-\frac{x}{2}\right)\right)} = \frac{1}{2} \cdot \frac{1}{1-\underbrace{\left(-\frac{x}{2}\right)}_{y}}$$

$$= \frac{1}{2} \sum_{n=0}^{\infty} \left(-\frac{x}{2}\right)^n = \frac{1}{2} \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{2^n} = \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{2^{n+1}}.$$

$\left|-\frac{x}{2}\right| < 1 \iff |x| < 2$. Hence

$$\frac{1}{2+x} = \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{2^{n+1}} \; ; \quad |x| < 2.$$

Example: Find a power series representation of $\frac{x^3}{x+2}$.

Sln:

$$\frac{x^3}{x+2} = x^3 \cdot \frac{1}{x+2} = x^3 \sum_{n=0}^{\infty} \frac{(-1)^n x^n}{2^{n+1}} = \sum_{n=0}^{\infty} \frac{(-1)^n x^3 x^n}{2^{n+1}}$$

$$= \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+3}}{2^{n+1}}.$$

$\text{Dom}(x^3) = \mathbb{R}$. $\text{Dom}\left(\frac{1}{x+2}\right) = (-2,2)$ (in the power series sense)

$\text{Dom}\left(\frac{x^3}{x+2}\right) = \text{Dom}(x^3) \cap \text{Dom}\left(\frac{1}{x+2}\right) = (-2,2)$

in the power series sense.

$$\frac{x^3}{x+2} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+3}}{2^{n+1}} \; ; \quad |x| < 2.$$

Differentiation and integration of Power Series.

Thm: If the power series $\sum_{n=0}^{\infty} C_n (x-a)^n$ has radius of convergence $R > 0$ then the function

$f(x) = \sum_{n=0}^{\infty} C_n (x-a)^n$ is differentiable (and therefore continuous) on the interval $(a-R, a+R)$ and

i). $$f'(x) = \frac{d}{dx}\left(\sum_{n=0}^{\infty} C_n (x-a)^n\right) = \sum_{n=0}^{\infty} C_n \frac{d}{dx}\left((x-a)^n\right) = \sum_{n=1}^{\infty} C_n \, n \cdot (x-a)^{n-1}$$

ii). $$\int f(x)\,dx = \int \left(\sum_{n=0}^{\infty} C_n (x-a)^n\right) dx = \sum_{n=0}^{\infty} C_n \int (x-a)^n dx$$

$$= \sum_{n=0}^{\infty} \frac{C_n}{n+1} (x-a)^{n+1} + C.$$

The radius of convergence of i). and ii). are both R.

Example: Express $\frac{1}{(1-x)^2}$ as a power series.

Sln:

$$\frac{1}{1-x} = \sum_{n=0}^{\infty}{}' x^n$$

$$\frac{d}{dx}\left(\frac{1}{1-x}\right) = \frac{d}{dx}\left(\sum_{n=0}^{\infty}{}' x^n\right)$$

$$\frac{1}{(1-x)^2} = \sum_{n=0}^{\infty}{}' \frac{d}{dx}(x^n) = \sum_{n=1}^{\infty}{}' n\,x^{n-1} = \sum_{n=0}^{\infty}{}' (n+1)\,x^n, \quad |x|<1$$

Example: Find a power series representation for $\ln(1+x)$ and its radius of convergence.

Sln: $$\frac{d}{dx}\left(\ln(1+x)\right) = \frac{1}{1+x} = \frac{1}{1-(-x)} = \sum_{n=0}^{\infty}{}' (-1)^n x^n$$

So,

$$\int\left(\frac{d}{dx}(\ln(1+x))\right)dx = \int\left(\sum_{n=0}^{\infty}{}' (-1)^n x^n\right)dx$$

$$\ln(1+x) = \sum_{n=0}^{\infty}{}' (-1)^n \int x^n dx = \sum_{n=0}^{\infty}{}' (-1)^n \frac{x^{n+1}}{n+1}$$

$$\ln(1+x) = \sum_{n=1}^{\infty}{}' (-1)^{n-1}\frac{x^n}{n}, \quad |x|<1$$

For you!

- Find a power series representation for $f(x) = \arctan x$.
- Evaluate $\int \frac{1}{1+x^7}\,dx$ as a power series.